

When Nothing Happens: Martingale-Coherent Distribution Forecasts for Prediction Markets

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Abstract

Prediction-market prices often remain unchanged for many observation periods and then move abruptly. Existing structural volatility models explain the size of active moves but do not provide an unconditional predictive distribution for the joint incidence and magnitude of price updates. We derive a restriction that couples these two margins. If a binary-claim price is a bounded martingale, the fraction of remaining Bernoulli uncertainty released over one period, $r_t = \text{Var}(p_{t+1} \mid \mathcal{F}_t) / [p_t(1 - p_t)]$, cannot exceed the update probability $q_t = \mathbb{P}(p_{t+1} \neq p_t \mid \mathcal{F}_t)$. We construct a martingale-coherent hurdle-beta distribution that realizes every feasible pair $0 < r_t < q_t \leq 1$, contains a point mass at no change, respects the $[0, 1]$ price boundary, and has tractable quantiles obtained from a standard beta inverse CDF. We also replace the locally additive DR-AS variance with a finite-step information-clock aggregation that cannot spend more than the remaining uncertainty and agrees with DR-AS to first order. A public-API replication evaluates the model in monthly expanding windows using proper mixed-distribution scores, multi-level coverage, and contract-cluster inference. Across 1,007,560 out-of-sample hours, the replication confirms the active-update DR-AS gain over deadline resolution. On the full panel MHB improves volume-weighted tick log score over no-hurdle and post-hoc-coherence ablations, with paired cluster intervals excluding zero, but it does not beat the active-fit DR-AS normal benchmark on volume-weighted log or interval score. The density advantage over DR-AS appears in Economics and reverses in Politics and Sports. The result identifies both the value and the empirical limit of martingale coherence without a scheduled-event or jump component.

1 Introduction

A prediction-market price is both a traded price and a probability. This dual role makes the standard volatility question unusually structured. The price is bounded, its terminal payoff is binary, and its resolution date is known. At the same time, the observed quote can remain exactly unchanged for hours even while uncertainty persists. A useful forecast must therefore answer two questions: will the displayed probability update, and how far can it move if it does?

Xi et al. (2026) derive a structural variance model with a Wright-Fisher deadline-resolution channel and a Glosten-Milgrom order-flow channel. Their DR-AS specification outperforms plain ARCH and GARCH models on a large Kalshi panel, especially after residual GARCH dynamics are added. The headline exercise deliberately conditions on active price updates. The paper's conclusion identifies a joint update-hazard and move-size model as a natural next step.

This paper develops that step as a distributional rather than merely regression extension. The central observation is that update incidence and conditional variance cannot be chosen independently once the price is required to remain a bounded martingale. Let q_t be the probability of any change and let r_t be unconditional one-period variance divided by remaining Bernoulli uncertainty.

We prove $r_t \leq q_t$. The inequality is sharp, easy to test, and economically interpretable: rare updates can release only a correspondingly limited share of uncertainty unless an active update is nearly a full resolution to zero or one.

The paper makes four contributions.

1. It gives a necessary and sharp coherence restriction linking the extensive margin of information arrival to the intensive margin of belief revision.
2. It constructs a hurdle-beta predictive law with exact martingale mean, exact unconditional variance, an explicit no-update atom, bounded support, and tractable central intervals.
3. It derives a finite-step information-clock version of DR-AS. The pure deadline component in fact has the exact normalized variance Δ/τ ; order-flow intensity enters through an exponential clock map that preserves the total variance budget and nests additive DR-AS locally.
4. It evaluates full predictive distributions on the unfiltered hourly panel using scores that are proper for mixed and price-lattice observations, rather than evaluating a variance proxy only after zero moves are removed.

The construction differs from general zero-inflated price models (Catania et al., 2019; Holý, 2023) because its two margins are coupled by the martingale and binary-payoff structure. It differs from logit jump-diffusion kernels for prediction markets (Dalen, 2025) because it targets the observed no-update atom and an unconditional short-horizon density, not a continuous-time derivative-pricing surface. Recent local-volatility work for event-market martingales (Xu, 2026) characterizes admissible boundary behavior and information clocks; the result here instead identifies a discrete-observation inequality between update incidence and variance release.

2 Structural benchmark and forecasting target

Let $\Theta \in \{0, 1\}$ be the settlement payoff at time T . Under the maintained risk-neutral interpretation,

$$p_t = \mathbb{E}[\Theta \mid \mathcal{F}_t] \quad (1)$$

is a bounded martingale with $p_T = \Theta$. For an observation step of length Δ and time to resolution $\tau = T - t$, the DR-AS benchmark of Xi et al. (2026) is

$$h_{\text{DR-AS}}^2 = \left[\frac{p_t(1-p_t)}{\tau} + K\nu(V_t)\frac{s_t^2}{4} \right] \Delta, \quad (2)$$

where s_t is the bid-ask spread, V_t is volume, ν is a concave activity transform, and $K \geq 0$ is fitted.

The benchmark maps h to a symmetric normal-reference interval around p_t and clips that interval to $[0, 1]$. This provides a common scoring rule for scale forecasts but is not the transition law of either structural channel. More importantly for the present paper, conditioning estimation and headline evaluation on $p_{t+\Delta} \neq p_t$ changes the variance object. We instead forecast the full law of $p_{t+\Delta}$ on the unfiltered panel.

3 A coherence law for update incidence

Fix a forecast origin and suppress time subscripts. Write $p \in (0, 1)$ for the current price, $Y \in [0, 1]$ for the next price, and

$$q = \mathbb{P}(Y \neq p \mid \mathcal{F}), \quad r = \frac{\text{Var}(Y \mid \mathcal{F})}{p(1-p)}. \quad (3)$$

Theorem 1 (Update-variance coherence). *If $Y \in [0, 1]$ almost surely and $\mathbb{E}[Y \mid \mathcal{F}] = p$, then*

$$0 \leq r \leq q \leq 1. \quad (4)$$

If $q > 0$, equality $r = q$ holds if and only if the active-update distribution is supported on $\{0, 1\}$ almost surely. If $r = 0$, then $q = 0$ under the definition $q = \mathbb{P}(Y \neq p \mid \mathcal{F})$.

Proof. Let $A = \{Y \neq p\}$. Since $Y - p = 0$ on A^c and $\mathbb{E}[Y - p \mid \mathcal{F}] = 0$, $\mathbb{E}[Y \mid A, \mathcal{F}] = p$ whenever $q > 0$. Hence

$$\text{Var}(Y \mid \mathcal{F}) = q\mathbb{E}[(Y - p)^2 \mid A, \mathcal{F}] \quad (5)$$

$$= q \text{Var}(Y \mid A, \mathcal{F}) \quad (6)$$

$$\leq qp(1 - p), \quad (7)$$

where the last line is the Bernoulli variance bound for a random variable in $[0, 1]$ with mean p : $\mathbb{E}[Y^2 \mid A, \mathcal{F}] \leq \mathbb{E}[Y \mid A, \mathcal{F}] = p$. Equality in that bound requires $Y^2 = Y$ almost surely conditional on A , which is equivalent to $Y \in \{0, 1\}$. If $r = 0$, then $Y = p$ almost surely, so $q = 0$. $\square$

The result does not rely on a diffusion, Gaussian shocks, or a particular market microstructure model. It is a consequence of three economically basic restrictions: bounded prices, the martingale mean, and an observed atom at no change. It also supplies a diagnostic. A separately estimated update hazard below a structural variance release cannot correspond to any bounded martingale transition with those moments.

4 Finite-step structural variance release

The neutral Wright-Fisher diffusion in information time u is

$$d\tilde{p}_u = \sqrt{\tilde{p}_u(1 - \tilde{p}_u)} dB_u. \quad (8)$$

Its generator applied to $g(p) = p(1 - p)$ gives $\mathcal{L}g = -g$. Therefore

$$\mathbb{E}[\tilde{p}_u(1 - \tilde{p}_u) \mid \tilde{p}_0 = p] = p(1 - p)e^{-u} \quad (9)$$

and, using the martingale mean,

$$\text{Var}(\tilde{p}_u \mid \tilde{p}_0 = p) = p(1 - p)(1 - e^{-u}). \quad (10)$$

The deadline clock has instantaneous rate $1/(T - t)$. Over a calendar step $[t, t + \Delta]$ with $\tau > \Delta$, its integrated information time is

$$\Lambda_{\text{DR}} = \int_t^{t+\Delta} \frac{ds}{T - s} = \log \left(\frac{\tau}{\tau - \Delta} \right). \quad (11)$$

Combining this expression with Equation (10) yields

$$r_{\text{DR}} = 1 - e^{-\Lambda_{\text{DR}}} = \frac{\Delta}{\tau}. \quad (12)$$

Thus, under the maintained Wright-Fisher clock, the pure deadline variance in Equation (2) is exact for a finite step ending before the deadline, even though it can also be obtained by an Euler argument.

To combine order flow without violating the remaining-uncertainty budget, define its variance-equivalent local clock intensity

$$\lambda_{\text{AS}} = K\nu(V_t) \frac{s_t^2}{4p_t(1-p_t)}. \quad (13)$$

We propose the finite-step release

$$r_t(K) = 1 - \exp\{-\Lambda_{\text{DR}} - \lambda_{\text{AS}}\Delta\}. \quad (14)$$

It lies in $[0, 1]$ by construction. A small-step expansion gives

$$r_t(K)p_t(1-p_t) = \left[\frac{p_t(1-p_t)}{\tau} + K\nu(V_t) \frac{s_t^2}{4} \right] \Delta + O(\Delta^2), \quad (15)$$

so the model nests DR-AS locally while introducing the finite-horizon interaction required by a single variance budget. The order-flow clock is a variance-equivalent reduced form; it is not claimed to turn the exact Glosten-Milgrom jump distribution into a Wright-Fisher diffusion.

5 A martingale-coherent hurdle distribution

For any $0 < r < q \leq 1$, set

$$\kappa = \frac{q}{r} - 1 \quad (16)$$

and define

$$Y \mid \mathcal{F} \sim (1-q)\delta_p + q \text{Beta}(p\kappa, (1-p)\kappa). \quad (17)$$

Proposition 1 (Sharp construction). *The law in Equation (17) has*

$$\mathbb{E}[Y \mid \mathcal{F}] = p, \quad \text{Var}(Y \mid \mathcal{F}) = rp(1-p), \quad \mathbb{P}(Y \neq p \mid \mathcal{F}) = q. \quad (18)$$

For $q = r$, its weak endpoint limit replaces the active beta component by a Bernoulli(p) law on $\{0, 1\}$. Together with the degenerate $r = q = 0$ case, the construction realizes every feasible pair in Theorem 1.

Proof. The active beta component has mean p and variance $p(1-p)/(\kappa+1) = (r/q)p(1-p)$. Mixture with a point mass at the same mean preserves the mean and multiplies active variance by q . A continuous beta draw equals p with probability zero, giving update probability q . As $\kappa \downarrow 0$, the beta law converges weakly to mass $1-p$ at zero and mass p at one. $\square$

The beta component is a parsimonious moment-matching kernel. It is not the exact finite-time Wright-Fisher transition density, which has nontrivial absorption behavior and is difficult to express analytically (Roa et al., 2022). This distinction is important: our theoretical claim concerns exact moments and the update atom, not exact recovery of the diffusion transition law.

We parameterize the hazard so coherence holds at every forecast origin:

$$q_t = r_t + (1-r_t) \text{logit}^{-1}(x'_t \gamma), \quad (19)$$

where x_t contains only forecast-origin states: log volume, log open interest, log time to resolution, log spread, $p_t(1-p_t)$, and the lagged update indicator. Equation (19) can be read as a structural floor r_t plus a residual probability of a partial, nonterminal quote update.

6 Estimation and proper evaluation

Each monthly test window is forecast using parameters estimated only on prior calendar months. We fit K on the full training panel by volume-weighted Gaussian quasi-likelihood for the unconditional variance $r_t(K)p_t(1 - p_t)$. Conditional on this release, γ is estimated by Bernoulli likelihood for the update indicator under Equation (19), using the same forecast-origin volume weights. This two-stage procedure targets the identifying moments directly and avoids using the future update indicator as a forecast-origin regressor.

The primary comparison contains five models:

1. deadline resolution with clipped normal-reference intervals;
2. the original active-update DR-AS fit with clipped normal-reference intervals;
3. the bounded beta kernel with $q = 1$, isolating distribution shape without a hurdle;
4. a separately fitted update hazard with post-hoc clipping to $q \geq r$;
5. the proposed MHB model, fitted with coherence enforced throughout.

Prices live on a market-specific lattice. Kalshi metadata identify cent, deci-cent, and tapered price-level structures; a mid-quote has local grid width equal to one-half the finest bid/ask tick at that observation. The primary log score therefore uses the forecast probability of the observed realized-price cell. For MHB this cell probability includes both integrated beta density and the no-update atom. For a clipped Gaussian it includes the normal probability mapped into the cell and the boundary mass created by clipping. This makes the score proper on a common discrete observation space and avoids assigning an arbitrary density unit to one model but not another.

We also report 50, 80, and 95 percent interval coverage, interval scores (Winkler, 1972; Gneiting and Raftery, 2007), update-hazard Brier scores, randomized probability integral transforms, and equal-, volume-, and contract-balanced aggregates. Contract-balanced weights normalize hourly volume to sum to one within each contract, limiting domination by a few high-volume claims. Contract-cluster resampling preserves within-contract dependence. All preprocessing thresholds and subgroup analyses are fixed before the full test outcomes are evaluated.

7 Public-API replication design

The independent replication uses Kalshi’s unauthenticated historical market and candlestick endpoints. Hourly candlesticks contain closing YES bid and ask, last price, volume, and open interest. The forecast price is the closing mid-quote. Market-level price ranges determine the score lattice. Only consecutive observations exactly one hour apart are retained. The scheduled expected-expiration field defines time to resolution; realized settlement timing is never used as a forecast-origin feature.

The frozen panel spans August 2021 through April 2026 and includes prespecified series from Economics, Climate and Weather, Politics, and Sports. Markets must have at least 48 valid one-hour forecast origins and at least 100 contracts of total volume. The main quote-quality rule requires a closing spread no wider than 20 cents; thresholds of 5, 10, 50, and 100 cents are sensitivity checks. Raw JSON is cached, every artifact is hashed, and failed requests are recorded in the run manifest.

This panel is an independent replication, not a claim of row-level identity with Xi et al. (2026). Historical API aggregation, market metadata updates, and quote cleaning may differ from the authors’ construction. The original headline ordering is treated as a reproduction target, but any difference is reported rather than silently tuned away.

8 Results

8.1 Locked out-of-sample results

The public-API acquisition selected 4,416 contracts and successfully retrieved 4,416. After forecast-validity, quote-quality, minimum-history, and expanding-window rules, the locked evaluation contains 1,007,560 hourly forecasts on 1,404 contracts across 50 test months. The unweighted update rate is 17.3%; its forecast-origin volume-weighted counterpart is 44.5%.

Table 1: Full-panel out-of-sample distribution forecasts. Lower is better for both scores.

Model	Tick log score	95% IS	Coverage	Width
DR normal	9.5984	0.2413	0.876	0.0971
DR-AS normal	3.1246	0.1896	0.980	0.1483
Beta, no hurdle	6.2704	0.2125	0.949	0.1286
Hurdle beta, post-hoc	6.0321	0.2148	0.949	0.1386
MHB	5.5271	0.2151	0.949	0.1390

On active updates, the independent replication preserves the source paper’s structural ordering: DR-AS posts a volume-weighted 95% interval score of 0.2744, versus 0.4412 for deadline resolution alone. The DR-minus-DR-AS improvement is 0.1667 with a 95% contract-cluster interval [0.0158, 0.2620].

Volume is concentrated: the two largest contracts contribute 50.3% of evaluation weight. Under the pre-specified contract-balanced robustness rule, MHB’s tick log score is 6.5503, versus 4.0972 for DR-AS normal and 8.8790 for the post-hoc hurdle; the corresponding interval scores are 0.2250, 0.2167, and 0.2246. Under equal observation weights, by contrast, MHB has the best tick log score (1.9974 versus 3.2032 for DR-AS) but a worse interval score (0.0787 versus 0.0750).

Table 2: Volume-weighted category results (descriptive).

Category	DR-AS log	MHB log	DR-AS IS	MHB IS
Economics	3.8702	3.4998	0.1857	0.1933
Politics	2.9638	3.4579	0.1589	0.1934
Sports	3.1669	9.7104	0.2435	0.2593

The MHB density advantage over DR-AS appears in Economics but reverses in Politics and Sports; interval scores do not improve. These category results are descriptive, not multiplicity-adjusted. The two pre-specified daily weather series contribute no contracts after the 48-valid-origin rule, which is a limitation of this public-API replication rather than a basis for changing the locked filter.

Against the active-fit DR-AS normal benchmark, MHB’s tick log score is higher by 2.4025 (95% contract-cluster interval [0.8307, 6.4983]). Against the active-fit DR-AS normal benchmark, MHB’s 95% interval score is higher by 0.0255 (95% contract-cluster interval [0.0107, 0.0374]).

Against the beta law without a hurdle, MHB’s tick log score is lower by 0.7433 (95% contract-cluster interval [0.6201, 1.0664]). Against the beta law without a hurdle, the beta law without a hurdle-minus-MHB 95% interval score difference is -0.0026; the 95% contract-cluster interval [-0.0145, 0.0045] includes zero.

Against the post-hoc coherent hurdle, MHB’s tick log score is lower by 0.5049 (95% contract-cluster interval [0.1329, 1.4202]). Against the post-hoc coherent hurdle, the post-hoc coherent

hurdle-minus-MHB 95% interval score difference is -0.0004; the 95% contract-cluster interval [-0.0011, 0.0000] includes zero.

Table 3: Volume-weighted scores by update regime.

Regime	Model	Tick log score	95% IS
All	DR-AS normal	3.1246	0.1896
	Beta, no hurdle	6.2704	0.2125
	Hurdle beta, post-hoc	6.0321	0.2148
	MHB	5.5271	0.2151
Active	DR-AS normal	3.9144	0.2744
	Beta, no hurdle	7.9759	0.3426
	Hurdle beta, post-hoc	9.4564	0.3378
	MHB	8.3244	0.3385
Inactive	DR-AS normal	2.4925	0.1218
	Beta, no hurdle	4.9054	0.1084
	Hurdle beta, post-hoc	3.2915	0.1164
	MHB	3.2884	0.1164

MHB central-interval coverage at the pre-specified levels is 50%: 67.2%, 80%: 87.7%, 95%: 94.9%. Its volume-weighted update-hazard Brier score is 0.2307, versus 0.2283 for the separately fitted hazard. The separate hazard violates $q \geq r$ at 5.49% of test origins before post-hoc repair.

Table 4: Pre-specified spread-threshold sensitivity.

Max spread	Post-hoc log	MHB log	Post-hoc IS	MHB IS
0.05	6.1602	5.5615	0.2168	0.2172
0.10	6.0445	5.5344	0.2151	0.2155
0.20	6.0321	5.5271	0.2148	0.2151
0.50	6.0268	5.5250	0.2149	0.2153
1.00	6.0608	5.5246	0.2151	0.2155

9 Limitations and interpretation

First, the displayed quote is not the frictionless efficient price. Exact staleness can reflect no information, thin liquidity, exchange closure, or a latent move smaller than one tick. The rounding-aware score addresses the price lattice but does not identify these mechanisms separately. Moreover, rounding a latent martingale need not itself produce a martingale. Theorem 1 is exact for the price process being modeled; its empirical use on displayed mid-quotes is therefore a maintained forecasting restriction, not an assertion that market microstructure preserves the martingale property exactly.

Second, the active beta law is a compact forecasting kernel rather than an exact transition density. It is useful because it is bounded, asymmetric, and moment coherent. Applications that require exact absorption probabilities should use a Wright-Fisher spectral or high-accuracy approximate transition and retain the same coherence law.

Third, a known resolution deadline is not the same as a known information event. Sports games, court decisions, and macro announcements concentrate information at scheduled or state-dependent

times. The current hazard absorbs some of that predictability through time, activity, and lagged updates, but a calendar of scheduled events is the natural next covariate set.

Finally, market prices equal risk-neutral rather than necessarily physical probabilities. The martingale result is a pricing-measure statement. Risk premia, fees, position limits, and manipulation can create predictable physical drift even if discounted prices are martingales under a selected measure.

10 Conclusion

In a bounded prediction market, “whether a price moves” and “how volatile it is” are not separable modeling choices. The martingale and binary-payoff structure imposes the sharp inequality $r \leq q$. Enforcing that law turns the DR-AS variance insight into an unconditional predictive distribution with a no-change atom, bounded support, and proper density forecasts. The locked evidence is deliberately narrower than a dominance claim. Coherence materially improves density fit over both hurdle ablations, yet the active-fit DR-AS normal benchmark remains stronger under the volume-weighted criterion, especially in Politics and Sports. Thus the theorem is a necessary design restriction, not a sufficient volatility model: event clocks, jump mixtures, or richer active-move kernels are required before the unconditional extension can dominate in event-heavy markets. That negative boundary is part of the contribution rather than a result to be tuned away.

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